2026 NASA SPACE APPS CHALLENGE · THE NEXT FRONTIER

PROHOR

Mission Health Intelligence

Your readings. Your baseline. A clearer next step.

An offline-capable health workspace for astronauts who must assess themselves,
with no clinic and a delayed link to the ground.

LIVE PROTOTYPE prohor-rho.vercel.app **SOURCE** github.com/tufayelahmedsuhagcc-dotcom/prohor

Challenge: Create Health Monitoring Software for Astronauts on Space Missions

Research prototype. Not a medical device. Independent project; no NASA endorsement implied.

Up to twenty-two minutes of delay. Five readings. No clinic.

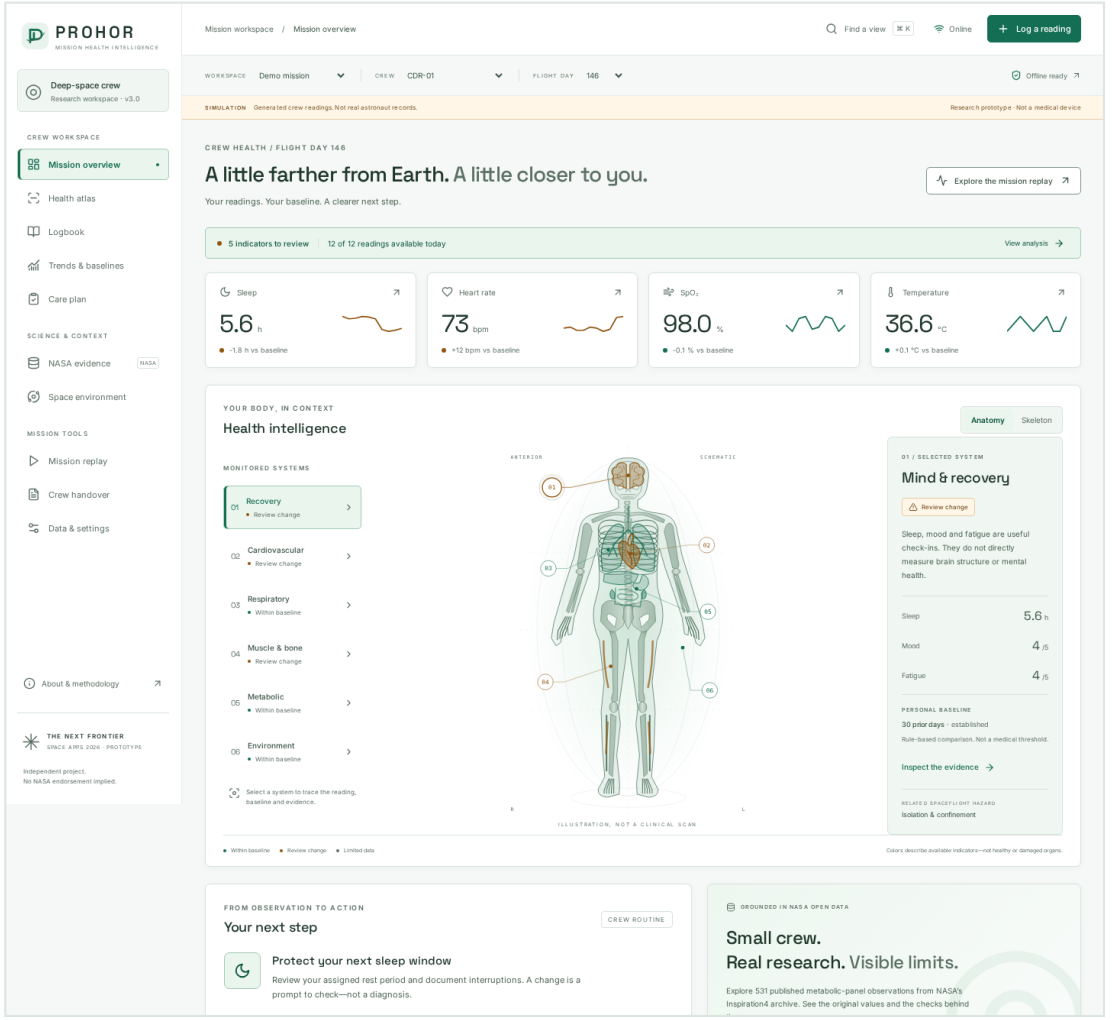
A crew member on a long-duration mission has to decide, alone, whether a number matters — and what to do next.

What did I measure?
A validated entry, in its original units.

How does it compare to my normal?
A baseline from this person's own prior days only.

How sure is this?
Coverage, direction and the rule, shown with the result.

What do I do now?
One conservative step, documented and handed to a surgeon.



Screenshot: mission overview, flight day 146 — five indicators under review.

02 / THE GAP

A colour is not a reason.

Most health dashboards answer with red, amber or green.

That answer hides the four things a crew member needs to judge it.

A flag with no baseline

Was this compared with the crew member, or with a textbook number?

A flag with no sample size

Two days of data is not a baseline. The interface rarely says which.

A flag with no rule

A threshold cannot be checked, argued with, or corrected.

A flag with no next step

An alarm without an action moves the decision back to the crew, unaided.

PROHOR's design rule: publish the uncertainty instead of a verdict.

Comparison you can audit.

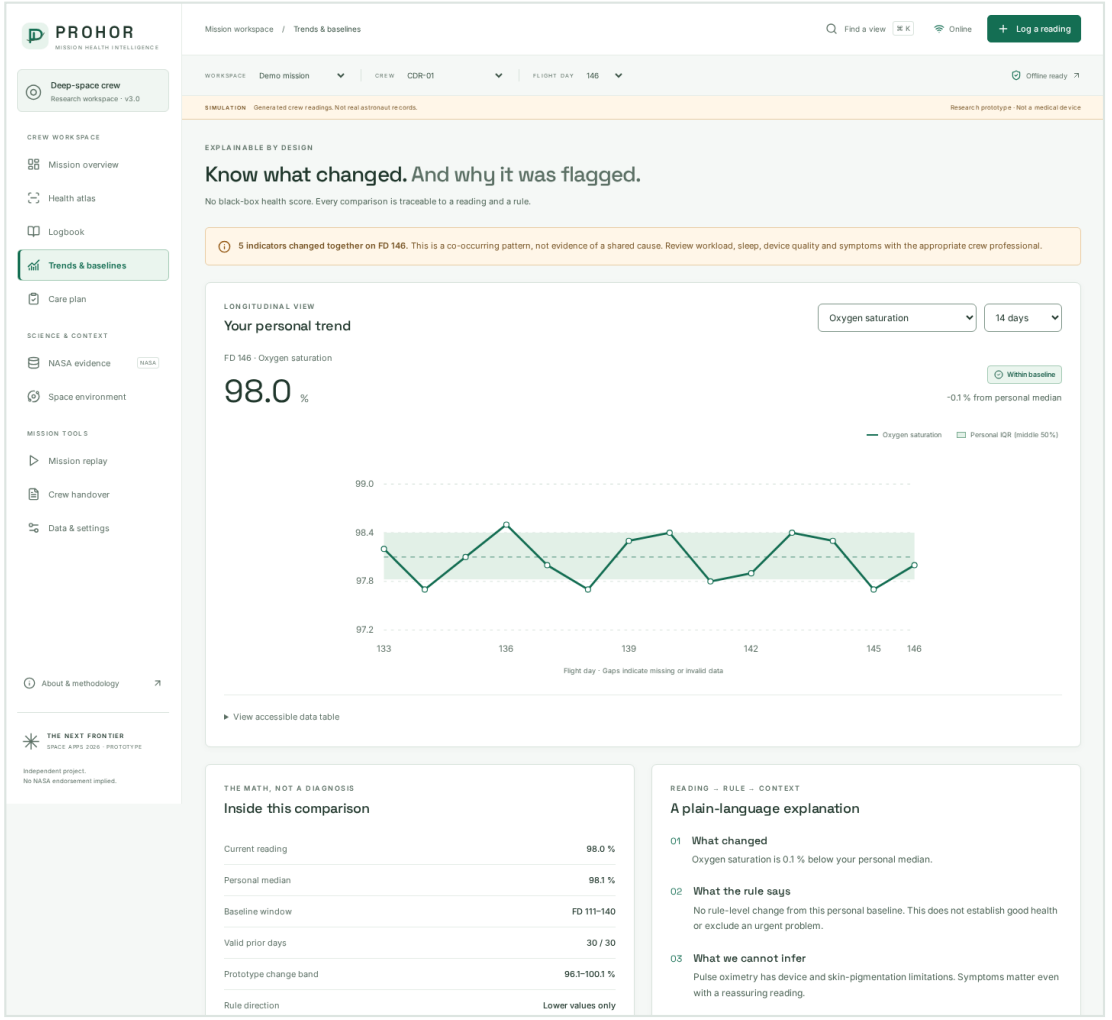
Median and interquartile range over the last 30 distinct prior days, excluding today, future days, invalid readings and missing values.

7 days minimum
below that, the app says so instead of reassuring you.

21+ days
labelled established — never a confidence score.

Never borrowed
no baseline from another crew member or another mode.

Every flag opens
window, day count, direction, source readings, table view.



Screenshot: trends & baselines — the rule, the band and the coverage on one screen.

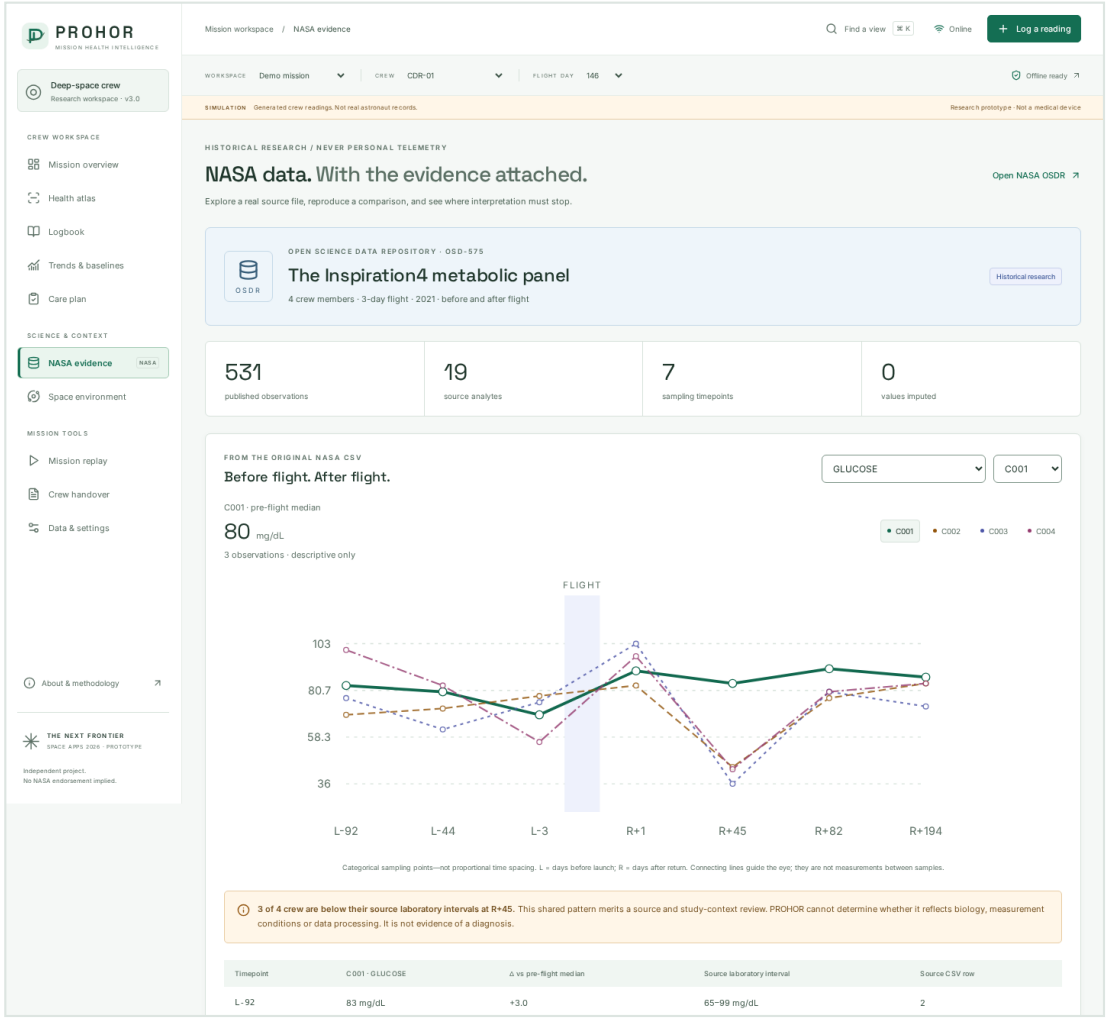
NASA data, checked in front of you.

OSD-575
531 published Inspiration4 metabolic-panel observations, parsed in-browser.

SHA-256
the original CSV bytes are hashed and verified on load.

Recomputed
medians recalculated per analyte and subject, with source row numbers.

Stated limits
four crew, one short flight, shared sample intervals — said plainly.



Screenshot: NASA evidence — glucose, subject C001, pre-flight median 80 mg/dL.

The packet, not the alert.

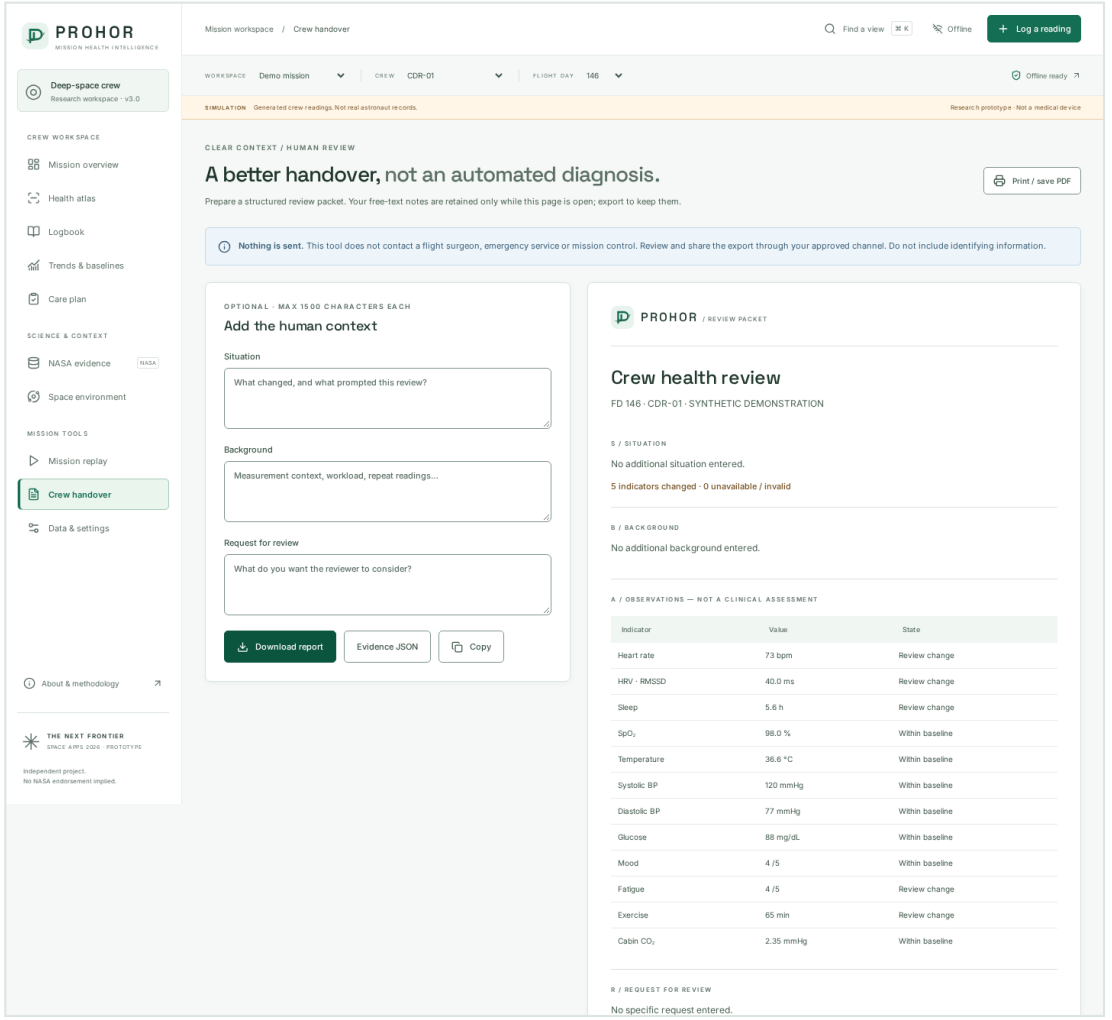
A flag ends in one conservative routine step, documented, and a structured handover the flight surgeon can read without re-deriving anything.

Care plan
done or deferred, persisted per crew, day and mode.

Handover
observations, baseline context, rule version, notes.

Export
text, JSON, clipboard or print-to-PDF. Nothing is transmitted.

Offline
installs from the browser and keeps working with no link.



Screenshot: crew handover — the review packet a surgeon receives.

What this is not.

Stated because it protects the crew it is designed for:

Not a medical device
no diagnosis, treatment or dosing, no dose estimate, no fitness call.

Not encrypted
single-user browser storage; no sync, accounts or multi-crew roles.

Not long-mission proof
the archive is four crew on one short flight.

NEXT

- Indicator set reviewed with flight surgeons
- On-orbit measurement ingestion
- Multi-crew roles and audited access
- Accessibility and clinical review

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SOURCE CODE github.com/tufayelahmedsuhagcc-dotcom/prohor

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